Continuous Ecological Monitoring
Methodology: from field audio to bird-activity metrics

How recordings become reliable detections, and detections become ecology

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The pipeline at a glance

Five stages. This strip reappears on every slide with the current stage filled in.

1

Record

Song Meter 2 at four stratified sites, 2 min on / 4 min off.

2

Pre-process

Segment, denoise, and flag rain before anything is detected.

3

Detect

BirdNET names the species in every 3-second window.

4

Filter

Five reliability steps turn raw detections into two clean datasets.

5

Analyse

Acoustic indices, activity regularity, habitat affinity, migration, solar alignment.

Notation used throughout

s
a bird species

j
a recording location (site)

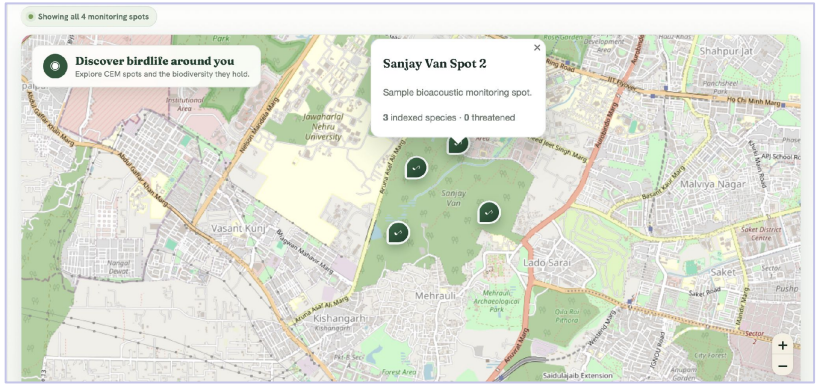
$k \in D = \{1, \ldots, n\}$
a day in the study period

$h \in H = \{1, \ldots, 24\}$
an hour of the day

$C_{s,j,k,h}$ the number of detections of species s at site j on day k during hour h. Every metric that follows is a sum or a correlation over this one quantity.

Study design

1 Record › 2 Pre-process › 3 Detect › 4 Filter › 5 Analyse



Sanjay Van, Delhi NCR: four stratified monitoring sites

Four stratified sites

Sites chosen to differ in vegetation composition, canopy structure and habitat, so comparisons across sites mean something.

Song Meter 2 recorders

Placed to maximise ambient sound capture and minimise anthropogenic noise.

2:4 duty cycle

2 minutes recording, 4 minutes sleep. Chosen from a pilot across sites: no measurable loss in species detection or index reliability, and longer field life.

Full equipment specifications, recording parameters and placement details are in the paper's appendix.

Pre-processing

1 Record › **2 Pre-process** › 3 Detect › 4 Filter › 5 Analyse

Three steps clean the audio before any species is named.

1 Segmentation

Each 2-minute file is one segment.

$T_{seg} = 120$ s, $f_s = 48{,}000$ Hz

2 Noise reduction

SNR-scaled noise subtraction in the time domain plus spectral gating in the frequency domain, applied through a short-time Fourier transform.

STFT → spectral gate → iSTFT

3 Rainfall removal

Heavy rain is broadband noise that inflates the Acoustic Diversity Index. A Random Forest on 40 MFCCs per 10-second chunk labels rain as None / Low / Medium / High; High segments are dropped before indices are computed.

94.17% overall accuracy

3,012 chunks from 251 files · 20% held-out test · Heavy class F1 = 0.96

Why it matters: without the rain filter, ADI produced false-positive biodiversity signals during the monsoon; with it, index values across the dataset were corrected.

Detection: BirdNET

1 Record › 2 Pre-process › **3 Detect** › 4 Filter › 5 Analyse

What BirdNET does

- An open-source classifier from Cornell and Chemnitz, trained on more than 6,000 species.
- Slides a 3-second window over each segment and returns, per window, a species and a confidence in [0, 1].
- Published weights are used unchanged; nothing is trained here.
- Every detection becomes one row in the project's aggregate.csv: species, confidence, file, site, date, hour.

What a detection is

$d \in D_s$ one detection of species s

$$c_{s,j,k,h} = \#\{\, d \in D_s : \text{site } j,\ \text{day } k,\ \text{hour } h \,\}$$

reads as how many times species s was heard at site j during hour h of day k. Everything downstream is built from this count.

Why confidence is a generation parameter

The confidence floor is set when BirdNET runs. A window below it is never written, so lowering the threshold later cannot recover it. Each site's effective floor is the maximum across its files, and it is recorded in the job log.

Filtering: five steps to a reliable table

1 Record › 2 Pre-process › 3 Detect › **4 Filter** › 5 Analyse

1 Taxonomic verification

keep only species on the regional eBird / Merlin checklist

$$S_{verified} = \{ s \in S_{raw} : s \in S_{eBird} \}$$

2 Confidence threshold

drop low-confidence windows

$$D_{filtered} = \{ d \in D_s : \mathrm{conf}(d) > 0.3 \}$$

3 Minimum activity

a species needs enough detections overall

$$C_s = \sum_j \sum_k \sum_h c_{s,j,k,h} > 10$$

4 Temporal consistency

and enough days with substantial activity

$$K_s = |\{ k : C_{s,k} > 10 \}| \geq 5$$

5 Rainfall removal

heavy-rain segments excluded before index computation

see Pre-processing

Two datasets

Quality-filtered

steps 1–3. The default for every analysis.

Temporally filtered

steps 1–4. Used where day-to-day structure matters: migration and solar alignment.

Where each is used

indices, regularity, affinity → quality-filtered
migration, solar → temporally filtered

C_s,k is the daily total for species s, summed over sites and hours.

Acoustic indices: the soundscape without species

1 Record › 2 Pre-process › 3 Detect › 4 Filter › **5 Analyse**

ADI — Acoustic Diversity Index

$$\mathrm{ADI} = -\sum_{i=1}^{N} p_i \log p_i, \quad p_i = \frac{E_i}{\sum_{j=1}^{N} E_j}$$

Shannon entropy of energy across N frequency bands. Higher = energy spread evenly, typical of many vocalising species.

ACI — Acoustic Complexity Index

$$\mathrm{ACI} = \frac{\sum_{t=1}^{T-1} |S(f,t+1) - S(f,t)|}{\sum_{t=1}^{T} S(f,t)}$$

Frame-to-frame change in amplitude, averaged over bins. High = rapid modulation, like birdsong; low = steady noise.

AEI — Acoustic Evenness Index

$$\mathrm{AEI} = 1 - G, \quad G = \frac{\sum_i \sum_j |E_i - E_j|}{2N \sum_i E_i}$$

Complement of the Gini coefficient of band energy. Penalises one band dominating.

NDSI — Normalized Difference Soundscape Index

$$\mathrm{NDSI} = \frac{E_{Bio} - E_{Anthro}}{E_{Bio} + E_{Anthro}}$$

Biophony (2–11 kHz) against anthrophony (1–2 kHz). −1 all traffic, +1 all birds and insects.

Urban masking. At Sites 2 and 3 a high AEI did not mean a balanced soundscape but white-noise saturation from distant traffic. NDSI is what penalises that; no single index is read alone.

Activity Regularity and Habitat Affinity

1 Record › 2 Pre-process › 3 Detect › 4 Filter › **5 Analyse**

The same idea twice: correlate consecutive days. AR asks when a bird calls; HA asks where.

Activity Regularity AR

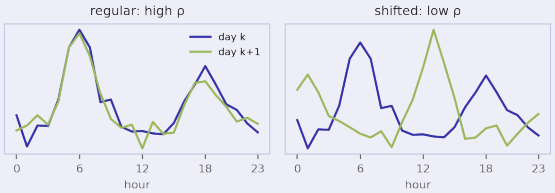
Does a species keep the same hourly schedule from one day to the next?

$$X_{s,j,k} = [\, c_1, c_2, \ldots, c_{24} \,] \quad \text{(hourly profile, day } k\text{)}$$

$$\rho_{s,j,k} = \text{Spearman}(X_{s,j,k},\, X_{s,j,k+1})$$

$$\text{AR}_s = \frac{1}{M_s} \sum_{j=1}^{M_s} \bar{\rho}_{s,j}, \qquad \bar{\rho}_{s,j} = \frac{1}{N_j} \sum_{k} \rho_{s,j,k}$$

pairs are used only when both days have at least one detection



Habitat Affinity HA

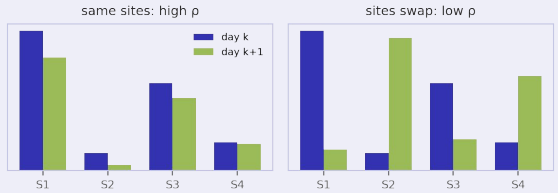
Does a species keep to the same sites from one day to the next?

$$Y_{s,k} = [\, c_{j_1}, c_{j_2}, \ldots, c_{j_m} \,] \quad \text{(per-site counts, day } k\text{)}$$

$$\rho_{s,k} = \text{Spearman}(Y_{s,k},\, Y_{s,k+1})$$

$$\text{HA}_s = \frac{1}{N_s} \sum_{k=1}^{n-1} \rho_{s,k}$$

pairs are used only when both days vary across sites



Both on the quality-filtered dataset. High values mean stable, predictable behaviour, a signature of resource-rich habitat.

Migratory classification

1 Record › 2 Pre-process › 3 Detect › 4 Filter › **5 Analyse**

Three complementary views of the daily series $C_{s,k} = \sum_{j}\sum_{h} C_{s,j,k,h}$ on the temporally filtered dataset.

SCI — Seasonal Concentration Index

$$\mathrm{SCI}_s = \frac{RS_{max,s}}{\sum_k C_{s,k}}$$

$$RS_{max,s} = \max_t \sum_{k=t}^{t+59} C_{s,k}$$

share of all detections inside the busiest 60-day window. Near 1 = migrant-like.

K — Residual kurtosis

$$K_s = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{r_i - \bar{r}}{\sigma}\right)^4$$

$$r_k = C_{s,k} - \hat{C}_{s,k} \quad \text{(after a linear trend)}$$

how peaked the residuals are after removing a linear trend. High = a few days of unusual bursts, like an arrival.

PMR — Peak-to-Median Ratio

$$\mathrm{PMR}_s = \frac{C_{max,s}}{\mathrm{Med}_s + \varepsilon}$$

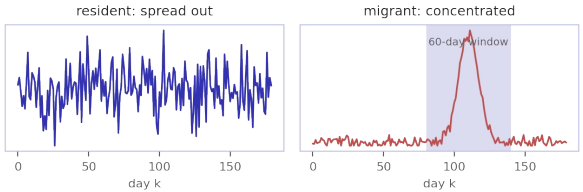
$$C_{max,s} = \max_k C_{s,k}, \quad \mathrm{Med}_s = \mathrm{median}_k\, C_{s,k}$$

the busiest day against a typical day. Large = extreme spikes relative to normal activity.

Decision rule

$$\text{migratory} \Leftrightarrow \mathrm{SCI}_s > \tau_{SCI} \ \wedge\ K_s > \tau_K \ \wedge\ \mathrm{PMR}_s > \tau_{PMR}$$

Thresholds set by inspecting species with known migratory behaviour, so that the criteria separate the two groups cleanly. Everything else is resident.



The silent resident. Coppersmith Barbet scored SCI = 0.97 because it only calls in May. eBird sightings showed it present but silent: a call-to-sighting check is needed before calling anything a migrant.

Solar alignment: do birds follow the light?

1 Record › 2 Pre-process › 3 Detect › 4 Filter › **5 Analyse**

Correlate each day's peak calling hour with that day's sunrise or sunset, on the temporally filtered dataset.

Definitions

Solar events, in decimal hours, from site coordinates (φ, λ):

$$SR_k = \text{Hour}_{sr} + \frac{\text{Minute}_{sr}}{60} + \frac{\text{Second}_{sr}}{3600}, \quad SS_k \text{ likewise}$$

Peak hour in each window, H_morning = {5…9}, H_evening = {17…20}:

$$P^{(m)}_{s,k} = \arg\max_{h \in H_{morning}} c_{s,h,k}, \qquad P^{(e)}_{s,k} = \arg\max_{h \in H_{evening}} c_{s,h,k}$$

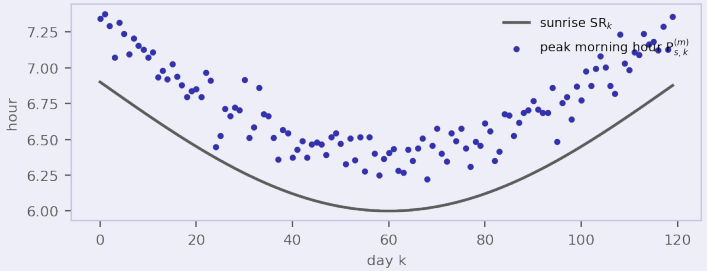
Lags, expected to stay constant if light drives the bird:

$$\Delta T_{sr} = P^{(m)}_{s,k} - SR_k, \qquad \Delta T_{ss} = P^{(e)}_{s,k} - SS_k$$

Correlation over days with a detection in that window:

$$\rho^{(m)}_s = \text{Pearson}(\{P^{(m)}_{s,k}\}, \{SR_k\}), \qquad \rho^{(e)}_s = \text{Pearson}(\{P^{(e)}_{s,k}\}, \{SS_k\})$$

Reading the result



High ρ(m) peak morning calling tracks sunrise as it shifts through the season: dawn drives the bird.

High ρ(e) the same for dusk.

Low ρ peak timing is set by something other than light: food, weather, competition.

Limitations and what they imply

Acoustics hears calls, not birds

A silent bird is invisible. Seasonal vocal patterns can make residents look migratory; eBird sightings are used as a cross-check.

The confidence floor is permanent

Detections below the run's threshold are never written. Sites processed at different thresholds have different floors, flagged in the index.

Timestamps come from filenames

Date and hour are read from SPOT_YYYYMMDD_HHMMSS names. A file that breaks the convention keeps its detections but loses its place in time.

Urban noise distorts indices

Traffic hum inflates AEI; rain inflates ADI. Both are handled (NDSI, the rain classifier) but neither correction is perfect.

Season length

SCI needs at least 60 days of recordings and the migration rule needs a full season; short deployments can only describe, not classify.

Thresholds are chosen, not learned

0.3 confidence, 10 detections, 5 days, and the migration τ values were set by inspection against known species, so they may need re-tuning for a new region.

Thank you